

Computational Photography Midterm

EEL 4403/5406

Fall 2026

Rules: This is a take home exam. You can consult a variety of sources of information, such your class notes, slides, websites, research papers, AI etc. Stick to the word limit of **100 WORDS** for each of the ten answers, but you can add **ONE figure** (if you like) per answer. The figure should not be bigger than a quarter of a page. Discussion with your groupmates is allowed. **I will only be available to talk about the exam during my office hours.** The point of this is to force you to think independently and so I will not reply to emails or phone calls (outside office hours) about the exam. This exam is due October 16th, 2026, by midnight. Late submissions will result in a bad grade. There are ten questions, each with ten points for a total of 100 points. I will only accept digital submission through Canvas.

You can use AI except where explicitly forbidden.

Every answer should be written by hand using a pencil/pen. I have a method to check for digitally created handwriting, which will result in an immediate F for the class. So please do not do it.

At the top of your submission, please type your full name and your major, including status (Grad, undergrad, senior, junior,...).

Q1. Consider a lamp with a bare bulb in an empty room with white walls. If the lamp is close to a wall, the wall appears brighter. If the lamp moves further from the wall, the wall is darker. This is due to the $1/R^2$ effect, or the inverse square fall-off effect. However, if the lamp moves further or closer to the camera, its size changes but its brightness remains the same. Why is there no inverse-square fall-off effect in the camera?

Q2. In class we looked at a paper about “Fast Separation of Direct and Global Images”. Google the title to look at the relevant paper and the web page. If you recall, the authors illuminated the scene twice, with two different patterns – a checkerboard and its inverse. Each pixel in the camera measured two images. When the scene point viewed by a pixel was illuminated by a white square (with intensity 1), the pixel measurements could be modeled as:

$$I_1 = I_{\text{direct}} + 0.5 I_{\text{global}}$$

When the scene point viewed by a pixel was illuminated by a black square (with intensity 0), the pixel measurements could be modeled as:

$$I_2 = 0.5 * I_{\text{global}}$$

- a) Write down equations for I_{direct} and I_{global} in terms of I_1 and I_2 .
- b) Recall that, for your direct-global lab, you used a linear occluder (i.e. a stick) to do direct-global separation. Clearly a line shadow is different from a checkerboard pattern. So, what are the equations for I_{direct} and I_{global} from your lab? You will need to estimate the percentage of the scene covered by the shadow which depends on the stick thickness.
- c) Your friend says that he can do the direct-global separation lab by keeping the stick static on a base and moving the light-source. Would this work or not? Why?

Q3: Remember photometric stereo? This was a technique for a static, Lambertian scene. The technique involved imaging with a single light source. Three (or more) images of the scene were collected by moving this single light source to different positions.

Photometric stereo works at each pixel independently. Consider one pixel, and let us only consider grayscale values (i.e. no color). If the p positions of the light source are 1, 2, 3, ... p , then the pixel measurements are

$$I_p = \rho * \text{dot}(\mathbf{n}, \mathbf{s}_p)$$

where ρ is a scalar (i.e. just a number) representing the grayscale color (i.e. albedo) of the scene, $\mathbf{n}=(n_x, n_y, n_z)$ is a unit vector representing the surface normal and $\mathbf{s}_p=(s_{x_p}, s_{y_p}, s_{z_p})$ are the p directions of the light source. *Dot* is the dot product function.

We discussed in class how to group together the I_p 's into a long vector $\mathbf{I}$, group all the $\mathbf{s}_p$'s into a matrix $\mathbf{S}$ and combine the ρ and the surface normal into a non-unit vector $\mathbf{n}'$, and use the equation

$$\mathbf{n}' = \text{pinv}(\mathbf{S}) * \mathbf{I} \quad , \text{ where } \text{pinv} \text{ is the pseudoinverse in Matlab.}$$

Given this review, answer the following questions. It may help you to type this up in Matlab, or you could do it by hand. To understand things better, it might be useful to first assume that $p = 3$ in all the following questions, and then think about what would happen with a larger p .

- a) Suppose the light-sources are near-light sources, which means that they have a fall-off effect. Also suppose they are point light sources, so they have an inverse-square fall-off effect. Rewrite the equation for I_p using the distance from the light-source to the scene point, which we will agree to depict by the letter R .

- b) How many unknowns per equation in the old equation for I_p ? How about the new one?
- c) What if the light source positions are not known, but the light-source is moving smoothly and you can get many images. How can you recover surface normals then? Do this for both inverse-square and non-inverse-square effect.

Q4. A friend of yours works for an underwater exploration company. They require reconstruction of underwater scenes. You realize the caustic-based underwater stereo reconstruction we learnt in class could be useful to them, especially if they commit to collecting data on a sunny day.

- a) You realize that the best place for good stereo reconstruction results is in shallow waters rather than deep waters. Why?
- b) Suppose your friend's needs reconstruction in deep waters. Design and describe a light-source that allows the algorithm for stereo reconstruction to remain the same as it was in shallow waters.
- c) Your friend now wants to do Dual Photography underwater at night (no caustics). A projector scans the ocean floor with a camera floating on a surface vehicle. Will this work if the water is murky?

Q5. In class we covered epipolar plane image (EPIs). Suppose someone gave you an epipolar plane image. Remember, it is made of lots of bands of lines at different slopes.

- a) A particular band of lines starts out with the same unique color in the first row of the EPI image, but in subsequent rows changes colors rapidly. Is the object Lambertian? Why or why not?
- b) If the camera moved in a parabola, what would be the form of the EPI? Write down its equation.

Q6. You have set up a projector and a camera to replicate some dual photography results. You place the projector on one side of a box full of interesting objects and the camera on top of the box. Once you generate the image from the projector's point of view, you notice that the image is much blurrier than you anticipated. What would cause the dual image to be blurred?

Q7. You are trying to create a depth edges image using the non-photorealistic camera algorithm, but some of the depth edges are missing in the result. When you examine your four images, and you notice the issue: the shadows are detached from the object in the scene. What can you do to fix this?

Q8. Describe one specific in-class lab where you might get better results using CLIP or DINO embeddings. Name the lab and white specific step the embeddings would replace or improve, and how would you use the embeddings to make your results better. Do not use an AI, because we will use one in the next question.

Q9. Discuss your idea from previous question with an AI. Ask it whether the idea is feasible and whether there are research papers that line up with it. Summarize the conversation in your own words. The summary must not be written by an AI. Attach the conversation as a share link, make sure the shared conversation contains no sensitive personal information.

Q10. In a previous lecture you were introduced to FoveaCam Duo. Each of its two foveas has an angular resolution of 502 px/° , and the two cameras are 200 mm apart. Assume a disparity error of pixel.

- a) Calculate its depth resolution, in mm, at standoff distances of 10 m, 50 m and 100 m.
- b) Repeat the calculation for two iPhone 16 Pro main cameras (48 MP, 8064 px wide, 24 mm-equivalent focal length) set up as a stereo pair with the same 200 mm baseline.
- c) Compare the two results, discuss your findings.

Show your derivation steps for part a and b.